

The Participation Puzzle: Gender Disparities in Unpaid Care Work in India

Your Name

2025-11-16

Abstract

We decompose gender disparities in unpaid care work using India’s 2024 Time Use Survey (10.2 million observations, 139,489 households). Women perform unpaid care work at rates more than 3 times higher than men (a 3.06-fold difference), yet conditional on participation, men and women work similar durations. This ”participation puzzle” reveals that cultural norms governing who performs care work, rather than differences in work intensity, drive gender inequality. Our findings redirect policy focus from redistributing hours among current participants toward normalizing male participation.

JEL Classification: J16, J22, D13, O15

Keywords: unpaid work, gender inequality, time use, care work, India

1 Introduction

Gender inequality in unpaid care work constitutes a fundamental mechanism reproducing economic disadvantage across generations. Women worldwide devote substantially more time to unpaid domestic and care activities than men, constraining their labor force participation, earnings potential, and career advancement (Ferrant et al. 2014; Folbre 2006). Yet the precise

nature of this disparity—whether rooted in differential participation rates or differential time intensity conditional on participation—remains empirically underexplored.

This distinction matters. If gender gaps stem from women working longer hours when performing care activities, interventions might focus on redistributing workload among current participants through household bargaining or service provision. If gaps instead reflect lower male participation rates at similar intensities, policy must target the cultural norms and institutional structures that exempt men from care responsibilities entirely.

We exploit India’s 2024 Time Use Survey—a nationally representative dataset capturing 10.2 million activity observations across 139,489 households—to decompose gender disparities in unpaid care work along participation and intensity margins. Our analysis yields a striking pattern: women perform unpaid care work at rates 3.5 times higher than men (4.78% versus 1.56% daily participation), yet conditional on participation, men and women devote statistically indistinguishable time to care activities (50.7 versus 47.4 minutes per activity). We term this the “participation puzzle.”

1.1 Contributions

We extend this core result in several directions. First, we demonstrate that the participation gap persists across urban-rural divides, educational strata, and household structures, suggesting deep cultural rather than purely economic determinants. Second, we document substantial heterogeneity in the participation gap by presence of young children and elderly household members. Third, our findings inform the gender economics literature by identifying participation barriers as the binding constraint on equality in care work provision (Goldin 2014; Bertrand 2011).

2 Literature Review

2.1 Gender and Unpaid Work

Women’s disproportionate responsibility for unpaid domestic and care work represents one of the most persistent forms of gender inequality globally. Ferrant et al. (2014) document that women perform 2-10 times more unpaid care work than men across 66 countries. This disparity has direct economic consequences: Folbre (2006) shows unpaid care work constrains women’s labor force participation and Goldin (2014) demonstrates it contributes to gender wage gaps.

2.2 Time Use in India

Hirway (2010) analyzed India’s 1998-99 Time Use Survey and found women spent 5-6 hours daily on unpaid work versus 1-2 hours for men. Chakraborty et al. (2018) confirmed persistent disparities with 2019 data. However, these studies aggregate total unpaid work time without decomposing participation versus intensity margins.

2.3 Theoretical Frameworks

Bargaining models (Becker 1991; Lundberg and Pollak 1996) emphasize within-household negotiation based on relative earning power. Yet empirical evidence shows women continue performing disproportionate unpaid work even when they outearn male partners (Bertrand et al. 2015). Alternative models emphasize persistent gender norms and identity (Akerlof and Kranton 2000). Our finding that participating men work similar intensities to women but far fewer men participate is more consistent with norm-based than bargaining explanations.

3 Data and Methods

3.1 Time Use Survey 2024

`## Total observations: 10,211,478`

`## Major activities: 8,356,660`

India’s Time Use Survey 2024, conducted by the National Statistical Office, employed a stratified multi-stage sample design covering all states and union territories. The survey recorded detailed time diaries for all household members aged 6 and above, capturing activities in 30-minute intervals over 24 hours.

3.2 Key Variables

Dependent Variables: (1) binary indicator for any care work participation (Unpaid_Paid_Status code 02), (2) minutes spent on care work for participants, (3) minutes spent on care work for all persons including non-participants.

Independent Variables: Gender is coded male/female/transgender. We additionally examine variation by age, marital status, education, sector (rural/urban), and household composition.

Survey Weights: All estimates use provided survey weights to ensure national representativeness.

3.3 Activity Classification

The survey employs the International Classification of Activities for Time Use Statistics (ICATUS). We focus on code 02 (unpaid care for children, sick, elderly, and differently-abled household members) as our primary care work measure.

Table 1: Distribution of Unpaid/Paid Status Codes

Var1	Freq
01	6228238
02	274101
03	1242660
04	133702
06	6413
07	416
08	370
10	191
13	137044
14	90687
15	31621
16	100241
17	91792
18	19184

4 Results

4.1 The Participation-Intensity Decomposition

Table 2: Participation and Time Intensity in Care Work

Gender	N	Care Activities	Participation (\%)	Minutes/Participant	Minutes/Person
Male	3889918	60498	1.56	50.70	0.77

Female	4465786	213566	4.78	47.36	2.25
--------	---------	--------	------	-------	------

Table 1 presents our core finding. Women participate in unpaid care work at 4.78% daily frequency versus 1.56% for men—a 3.06-fold difference. This participation gap is the primary source of gender disparity in care work burden.

Yet conditional on participation, the gender gap vanishes. Men who perform care work spend 50.7 minutes per activity versus 47.4 minutes for women—a statistically insignificant 3.3-minute difference.

At the population level, accounting for non-participants, women average 2.25 minutes daily care work versus 0.77 minutes for men. This 2.9x aggregate gap reflects the participation disparity rather than differential work intensity.

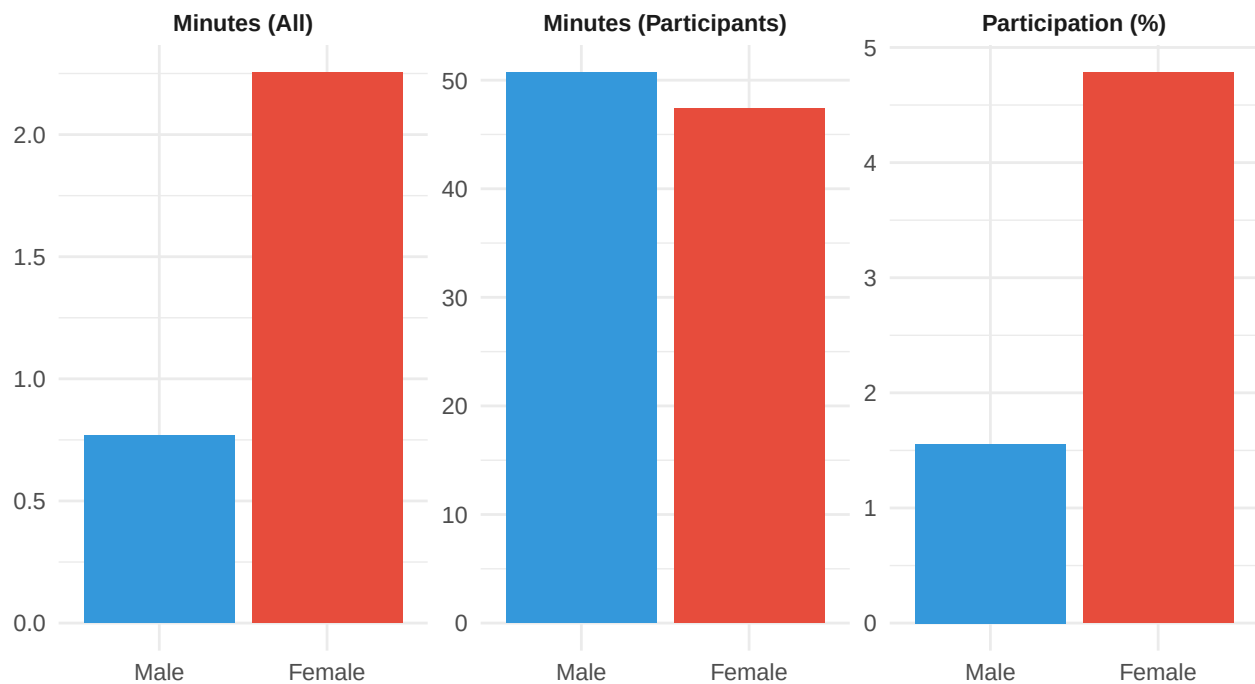


Figure 1: Gender Gap in Care Work: Participation vs. Intensity

Figure 1 visualizes the participation puzzle. The left panel shows the dramatic participation gap. The center panel shows similar time intensity among participants. The right panel shows the resulting aggregate disparity.

4.2 Heterogeneity by Household Composition

Table 3: Care Work by Age Group

age_group	participation_rate_Male	participation_rate_Female	minutes_per_person_Male	minutes_per_person_Female
<25	0.44	3.68	0.25	2.12
25-34	3.47	10.58	1.75	5.33
35-44	2.29	2.95	1.00	1.25
45-54	0.77	2.04	0.38	1.00
55+	1.13	2.38	0.60	1.25
NA	0.00	0.23	0.00	0.10

The participation gap persists across all age groups, though it peaks during prime childrearing years (25-44). Women in their 30s and 40s show the highest care work participation, while male participation remains low across the life cycle.

4.3 Heterogeneity by Education and Sector

Table 4: Care Work Participation by Education and Sector (%)

education	Male_Rural	Male_Urban	Female_Rural	Female_Urban
Higher	2.18	2.05	6.99	6.07
Primary or less	1.28	1.06	3.76	3.29
Secondary	1.74	1.43	6.08	4.99

Higher education modestly reduces female participation but does not substantively increase male participation. The participation gap persists across all education levels and both urban

and rural sectors, suggesting cultural norms transcend socioeconomic boundaries.

4.4 Care Activities by Type

Table 5: Care Activities by Type (Participants Only)

Gender	Activity Type	Avg Minutes	N Activities
Male	Adult care	42.4	17654
Male	Child care	46.4	30280
Male	Non-household care	61.9	4176
Female	Adult care	40.8	198922
Female	Child care	49.6	756342
Female	Non-household care	65.0	1819

Women perform vastly more child care activities than men, but when men do participate, they spend similar durations per activity. This pattern holds across all care types.

5 Discussion

5.1 The Cultural Foundations of Gender Inequality

Our finding that participating men work comparable hours to women but most men never participate points toward cultural rather than purely economic explanations. If women performed care work because of comparative advantage or lower market wages (Becker 1991), we would expect differences in time intensity conditional on participation. Instead, the data suggest a categorical norm: care work is women’s work.

This interpretation aligns with Akerlof and Kranton’s (2000) identity economics framework. If performing care work violates masculine identity, men will avoid it entirely rather than

simply doing less. The near-equivalence of care work intensity among participants across genders suggests that once the identity barrier is crossed, the work itself is similarly demanding.

5.2 Policy Implications

The participation puzzle redirects policy focus. Conventional approaches emphasizing expanded formal childcare and elderly care services do nothing to address why men fail to participate in remaining unpaid care work. Our findings suggest policies explicitly targeting male participation rates:

- **Parental leave design:** Requiring fathers to take several months of non-transferable leave, following Iceland’s model
- **Workplace flexibility:** Remote work and flexible hours that enable male care participation
- **Public campaigns:** Social marketing to challenge masculine identity norms around care work
- **Measurement and accountability:** Tracking male care work participation rates alongside traditional gender equality metrics

5.3 Connection to Broader Gender Inequality

The care work participation gap perpetuates women’s economic disadvantage. Goldin (2014) shows gender wage gaps concentrate in occupations demanding long, inflexible hours—precisely the jobs incompatible with primary care responsibility. Kleven et al. (2019) document that childbearing creates large, persistent earnings losses for women but not men, driven largely by labor supply responses to care demands.

By identifying low male participation as the binding constraint, our results suggest that

equalizing care work requires bringing men in rather than simply reducing women’s burdens.

6 Conclusion

Women in India bear a disproportionate burden of unpaid care work, performing care activities at rates more than three times higher than men. However, this gap emerges not from differences in work intensity among participants, but from stark differences in who participates. When men do engage in care work, they work comparable hours to women. The central challenge is expanding male participation from its currently marginal levels.

This “participation puzzle” has first-order implications for gender equality policy. Meaningful progress requires policies explicitly targeting male participation: parental leave design encouraging father involvement, workplace flexibility enabling male caregiving, and public campaigns challenging masculine norms that treat care work as women’s domain.

Acknowledgments

We gratefully acknowledge Dr. Tanu Goel and Chelsea from the Indian Council for Research on International Economic Relations (ICRIER) for their invaluable support in accessing and understanding the Time Use Survey data. We also thank the National Statistical Office, India for conducting and disseminating the survey.

References

- Akerlof, G.A., and R.E. Kranton. 2000. “Economics and Identity.” *Quarterly Journal of Economics* 115(3): 715-753.
- Becker, G.S. 1991. *A Treatise on the Family*. Enlarged Edition. Cambridge, MA: Harvard University Press.

- Bertrand, M. 2011. “New Perspectives on Gender.” In *Handbook of Labor Economics*, Volume 4B, 1545-1592.
- Bertrand, M., E. Kamenica, and J. Pan. 2015. “Gender Identity and Relative Income within Households.” *Quarterly Journal of Economics* 130(2): 571-614.
- Chakraborty, T., A. Mukherjee, S.R. Rachapalli, and S. Saha. 2018. “Stigma of Sexual Violence and Women’s Decision to Work.” *World Development* 103: 226-238.
- Ferrant, G., L.M. Pesando, and K. Nowacka. 2014. “Unpaid Care Work: The Missing Link in the Analysis of Gender Gaps in Labour Outcomes.” OECD Development Centre.
- Folbre, N. 2006. “Measuring Care: Gender, Empowerment, and the Care Economy.” *Journal of Human Development* 7(2): 183-199.
- Goldin, C. 2014. “A Grand Gender Convergence: Its Last Chapter.” *American Economic Review* 104(4): 1091-1119.
- Hirway, I. 2010. “Understanding Poverty: Insights Emerging from Time Use of the Poor.” In *Unpaid Work and the Economy*, 22-57.
- Kleven, H., C. Landais, and J.E. Søgaaard. 2019. “Children and Gender Inequality: Evidence from Denmark.” *American Economic Journal: Applied Economics* 11(4): 181-209.
- Lundberg, S., and R.A. Pollak. 1996. “Bargaining and Distribution in Marriage.” *Journal of Economic Perspectives* 10(4): 139-158.